Data mining

18/11/2024

Model evaluation and regulation

In supervised learning the aim is to infer relevant information from given data, to parameterize it in terms of a model , and to apply it to novel data successfully.

It is essential to know or at least have some estimate of the performance that can be expected in the prediction phase.

Different sources of error can influence the performance of supervised learning systems:

- Bias
- Variance

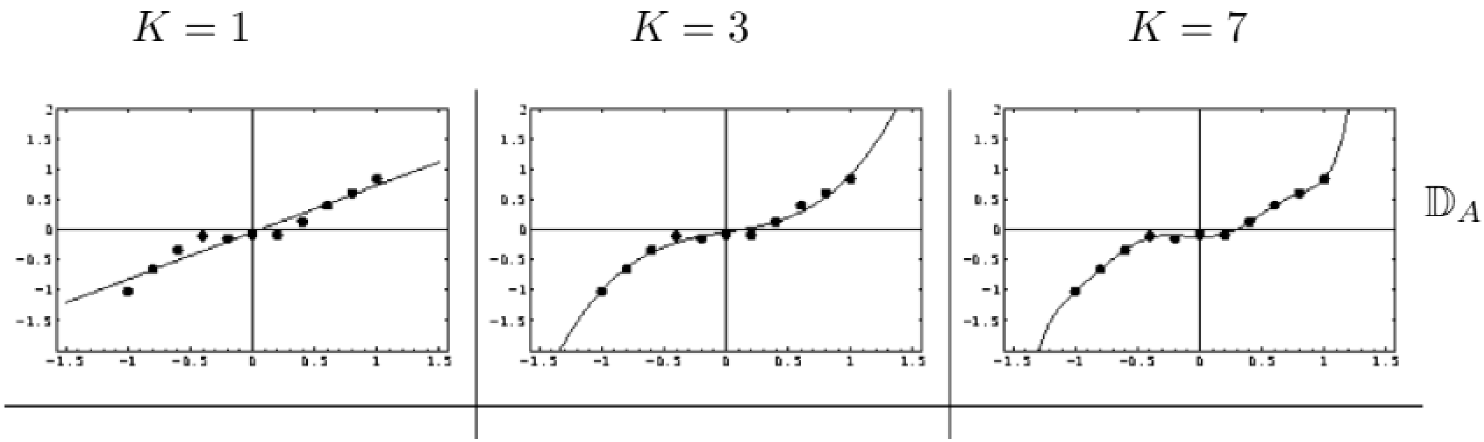
Decomposition in Bias and Variance

The expected prediction error can be decomposed into:

- Bias
  - Systematic deviation of the trained model from the true target.
  - Error from erroneous assumptions in the learning algorithm.
- Variance
  - Variations of the model performance when trained from different training data.
  - Sensitivity to small fluctuations in the training set.

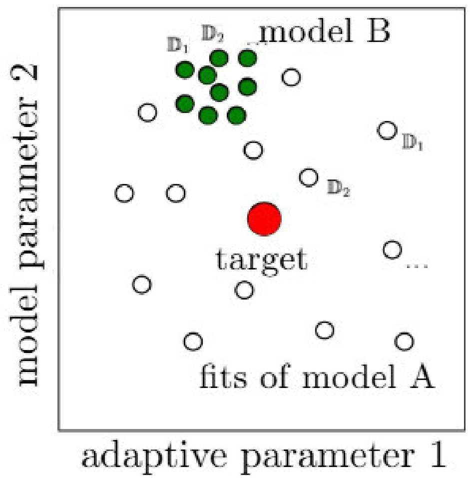
Bias

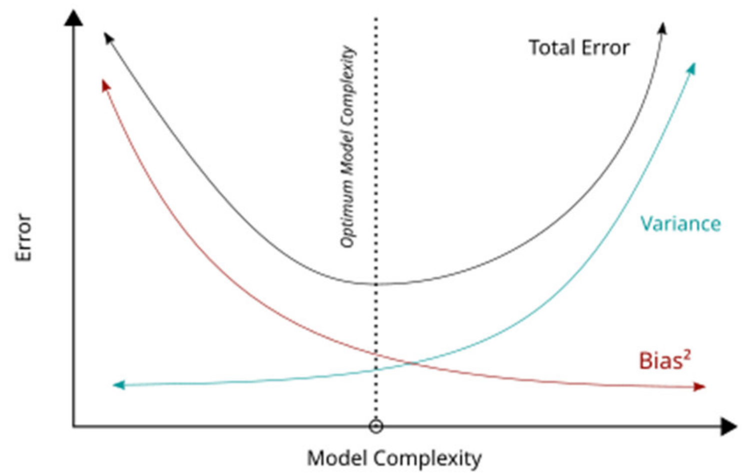
- Bias (systematic deviation of the trained model from the true target).
- llustration of the bias in regression according to model complexity:



Bias-Variance dilemma

Goal: Achieve low variance and low bias at the same time, i.e. a robust and faithful approximation of the target rule.





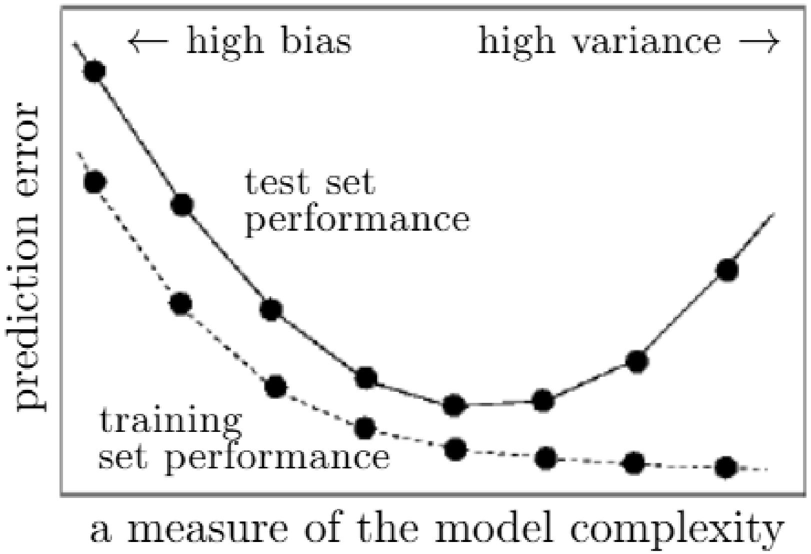
Overfitting and underfitting

Overfitting

- Low bias – High variance
- Unable to generalize well for unseen data.
- Risk to model the random noise in the data.

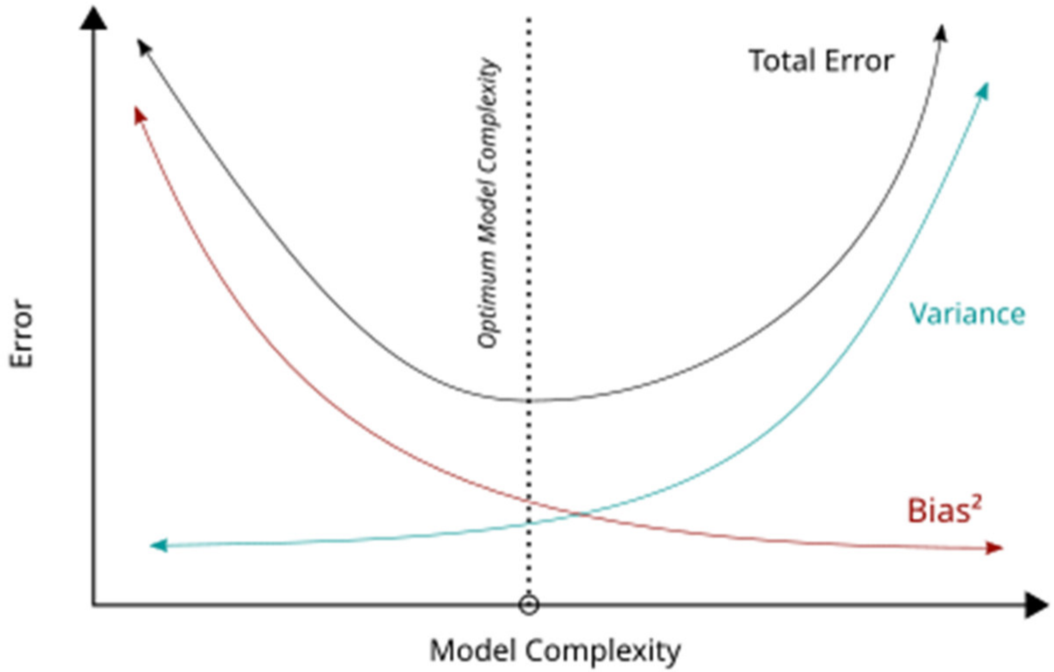
Underfitting:

- Low variance – High bias
- Unable to capture relevant relations between and features and target outputs.



Bias-Variance dilemma

Goal: Achieve low variance and low bias at the same time, i.e. a robust and faithful approximation of the target rule.



Comparison methods

High-variance learning methods may be able to represent their training set well but are at risk of overfitting to noisy or unrepresentative training data.

In contrast, algorithms with high bias typically produce simpler models that may fail to capture important regularities (i.e. underfit) in the data.

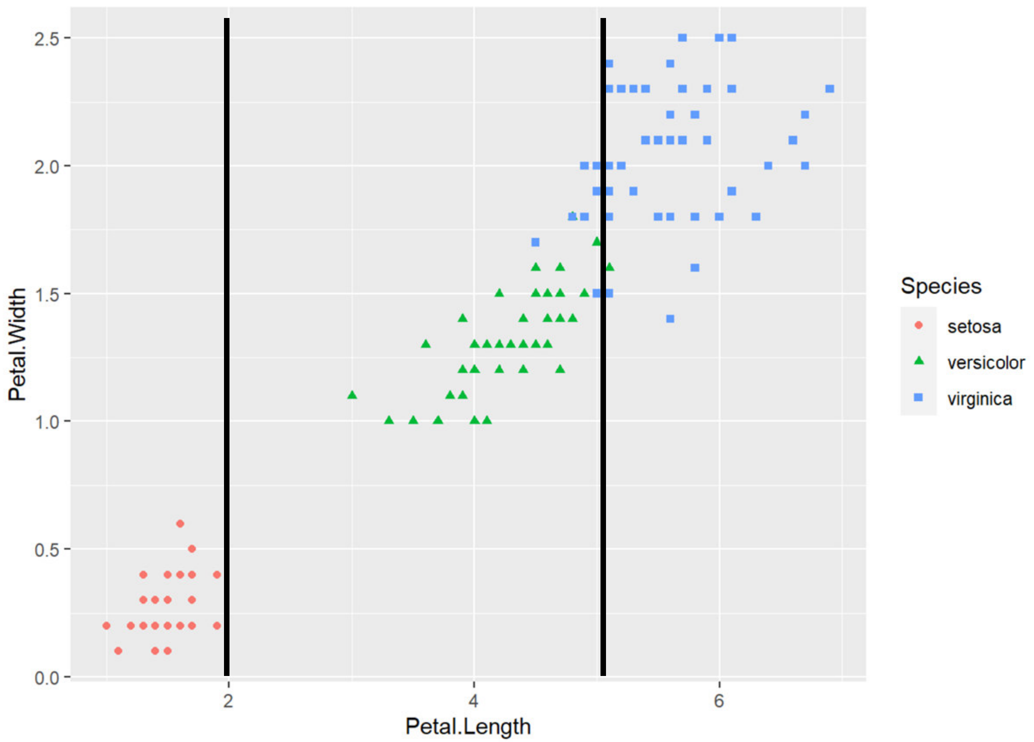
Control Bias and Variance

Learning algorithms typically have some tunable parameters that control bias and variance.

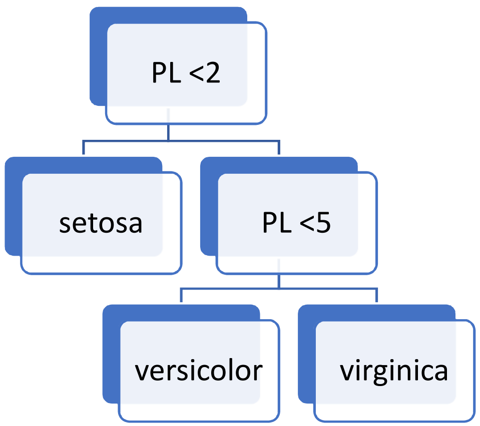
- K-Nearest Neighbor, a high k leads to high bias and low variance.
- Decision Trees, the depth of the tree determines the variance (Pruning).
- Neural Networks, an increase of the number hidden units increases variance and decreases bias.

Model validation (cross-validation) can be used to tune models and to optimize the bias-variance tradeoff.

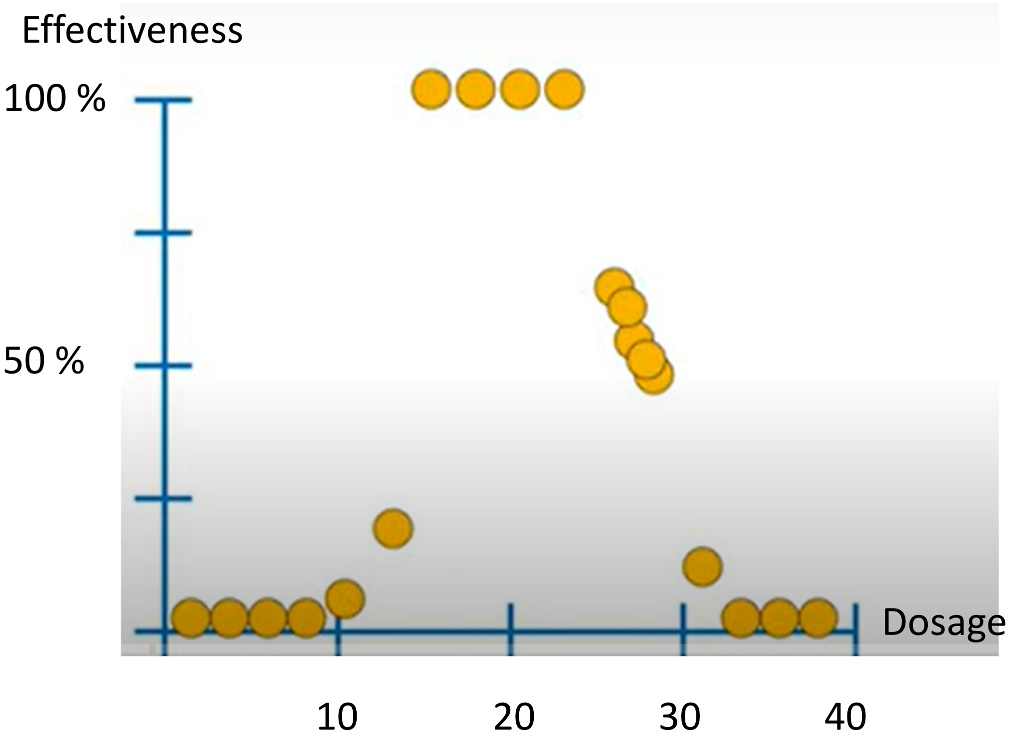
Decision Trees for classification

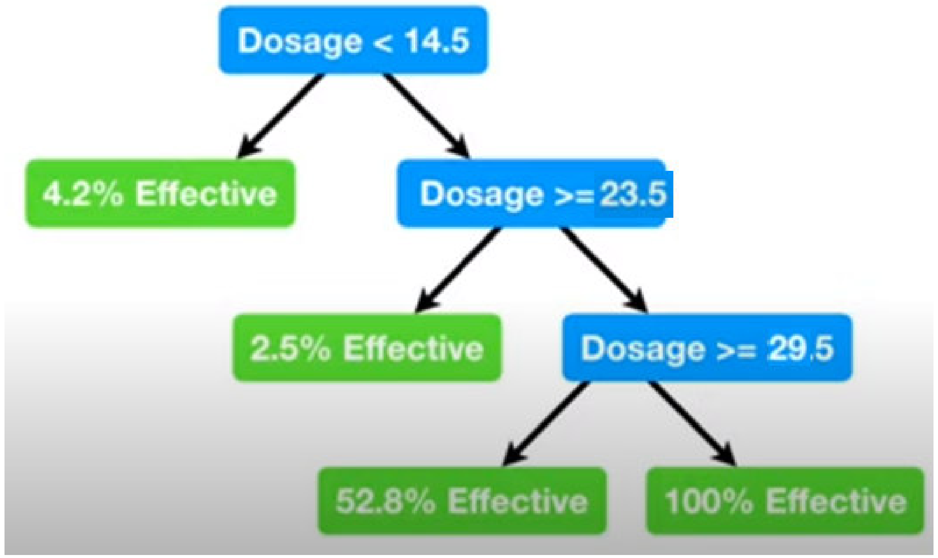


$$IG(\mathcal{X}, A) = E(\mathcal{X}) - E(\mathcal{X}|A)$$

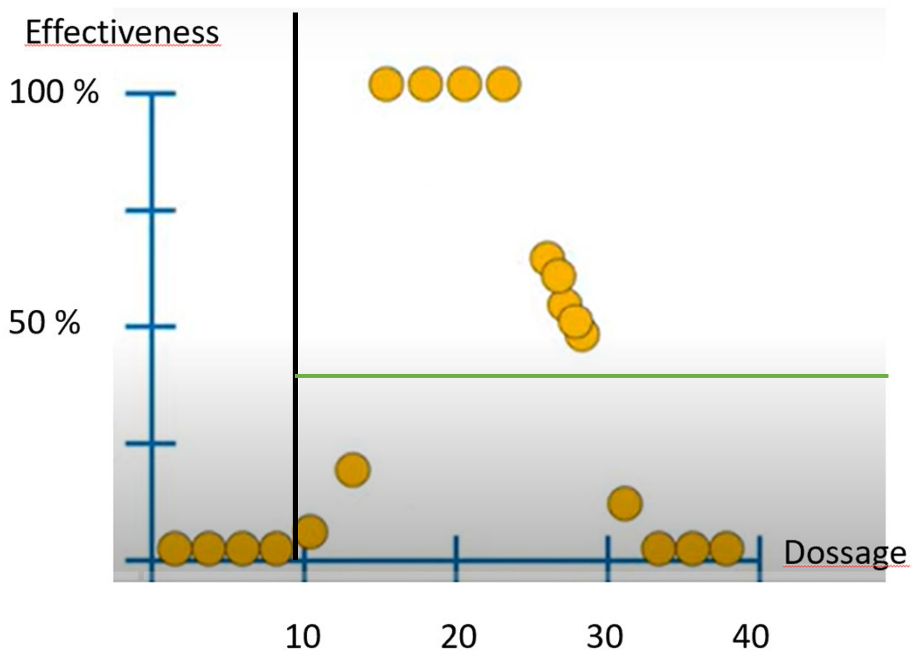


Regression Tree



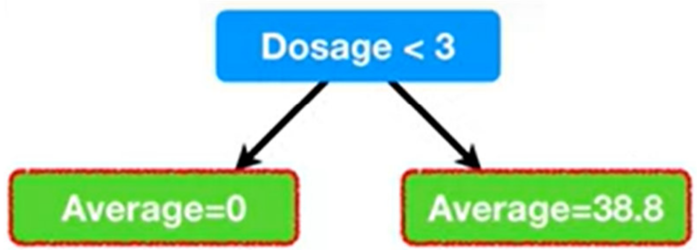


Find boundaries

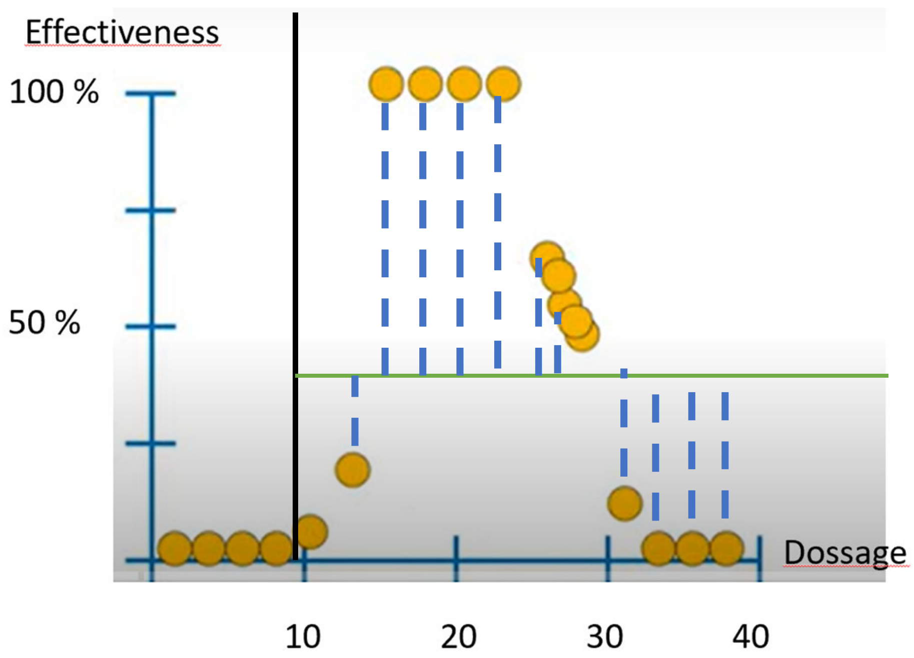


Calculate Sum of Squared Residuals (SSR):

SSR = $\sum_{i=1}^{i=n} vi - mean$

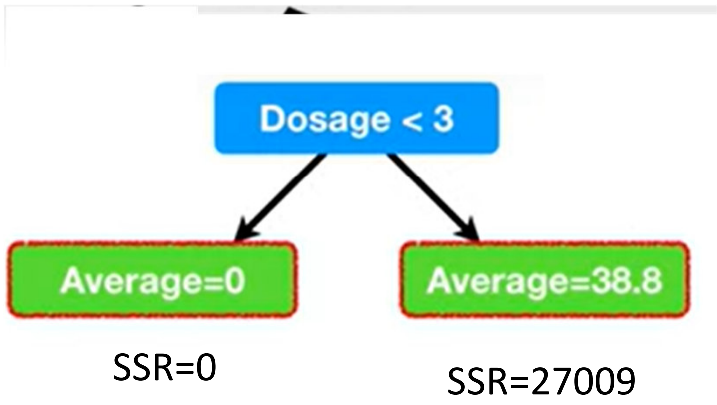


Sum of squared residuals

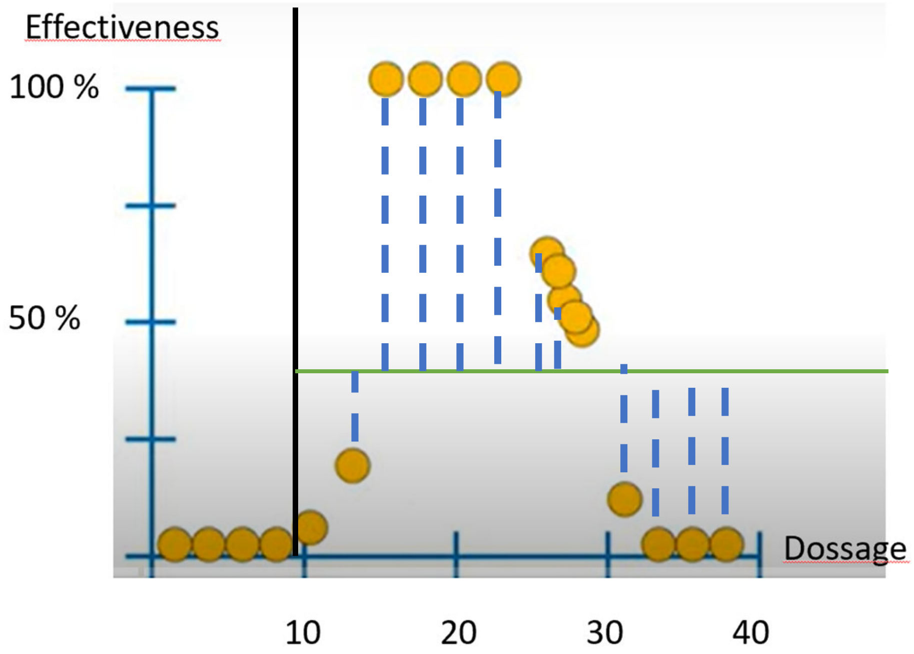


Calculate SSR:

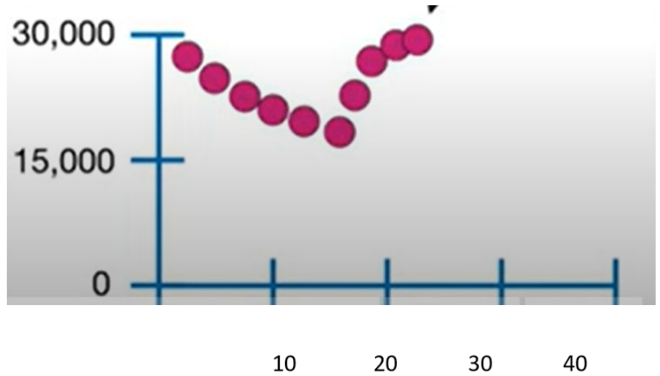
SSR = $\sum_{i=1}^{i=n} vi - mean$



Optimizing boundaries



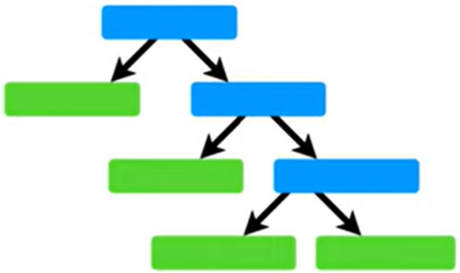
SSR = $\sum_{i=1}^{i=n} vi - mean$



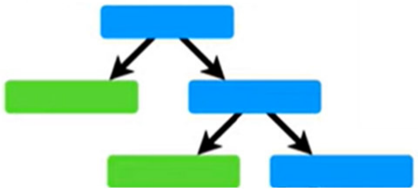
Quality of trees

- Tree Score= SSR
- Leads to complex trees and overfitting

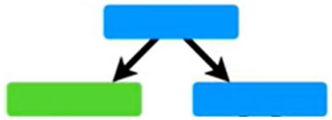
SSR= 2.500



SSR= 21.500



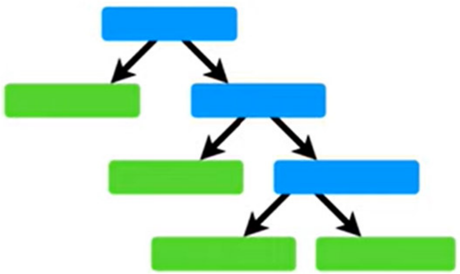
SSR= 110.500



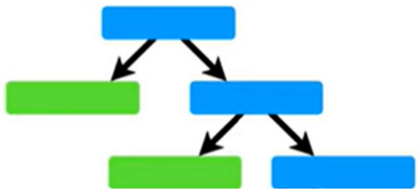
Alternative: Tree complexity penalty

- Controls the complexity of the tree by giving a penalty P to the number of nodes or levels.
- Tree Score= SSR + αP

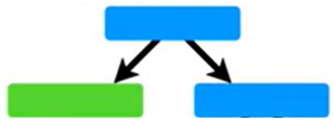
SSR= 2.500



SSR= 21.500



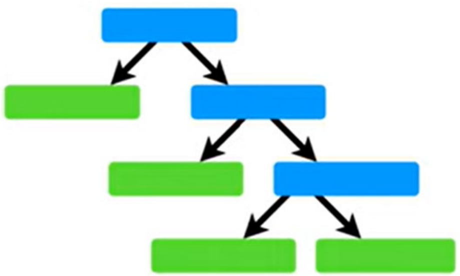
SSR= 110.500



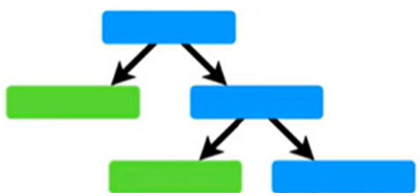
Alternative: Tree complexity penalty

- Controls the complexity of the tree by giving a penalty P to the number of nodes or levels.
- Tree Score= SSR + 20.000* levels

TR= 60.000 + 2.500 = 62.500
SSR= 2.500



TR= 61.500
SSR= 21.500



TR= 130.500
SSR= 110.500